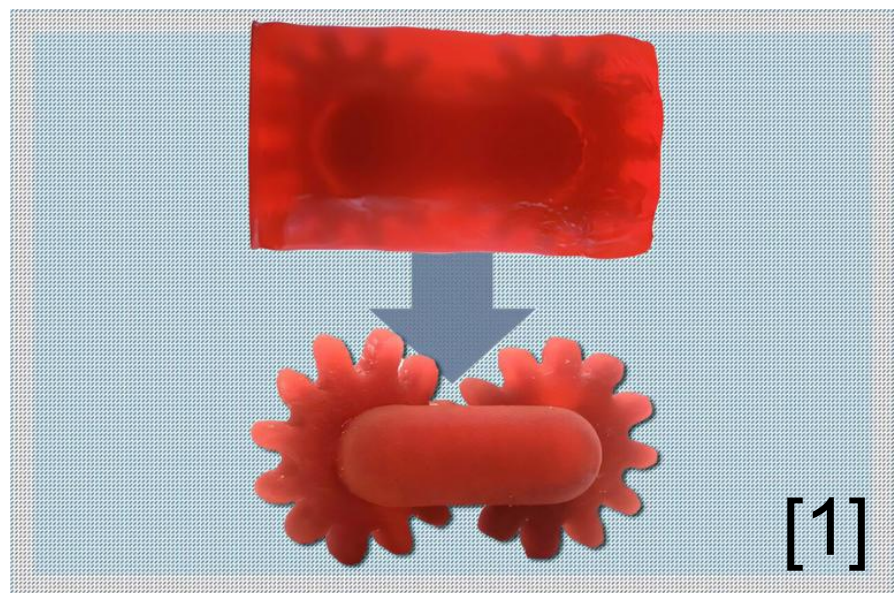




Shelf Life of Novel Resin for DLP 3D Printing



Katherine Jankowski

2.671 Measurement and Instrumentation

Abstract

It is important to know the shelf life of resins used for DLP 3D printing in order to effectively make quality parts and reduce waste. Researchers at MIT have developed a resin capable of producing parts with soluble supports, and the stability of this resin is not yet known. This study investigated the stability of this novel resin through Gas Chromatography Mass Spectrometry measurements of pyridine, an important component of the resin which is hypothesized to decay in the resin over time. Results of GC-MS measurements on 3 different resin proxies with two different preparation methods showed that pyridine levels did not change significantly over a 9-day aging period. Thus, pyridine is likely not the chemical responsible for the shelf life of the resin, and further investigation of the resin’s stability will need to be done.

Experimental Design

Sample
Mixing

Aging

Dilution for
GC-MS

GC-MS

Samples:

Control Mixture

BAPO + Pyridine

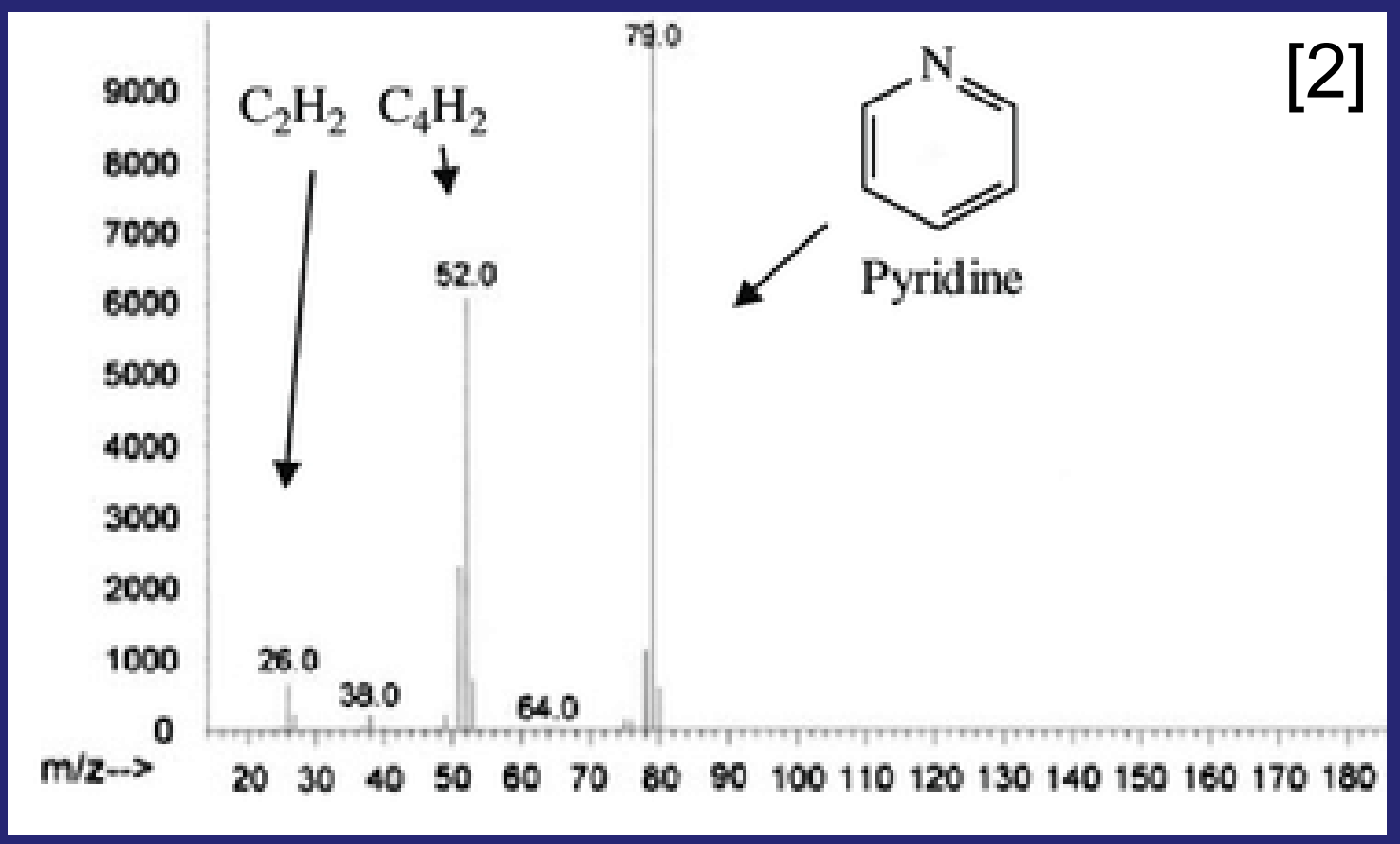
Full Resin Proxy

Control Mixture (aged
after dilution)

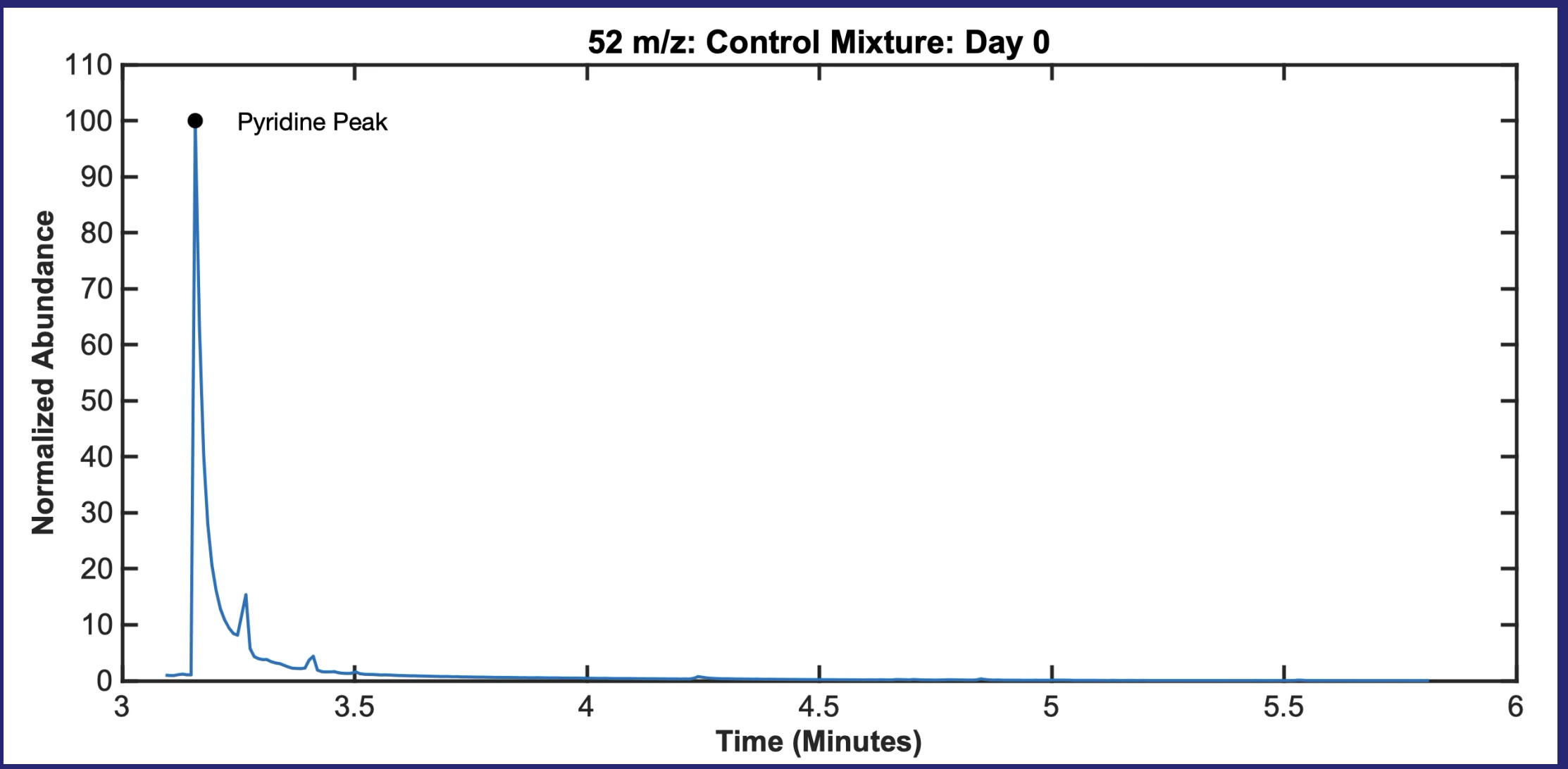
BAPO + Pyridine
(aged after dilution)

Full Resin Proxy (aged
after dilution)

Pyridine Mass Spectrum:



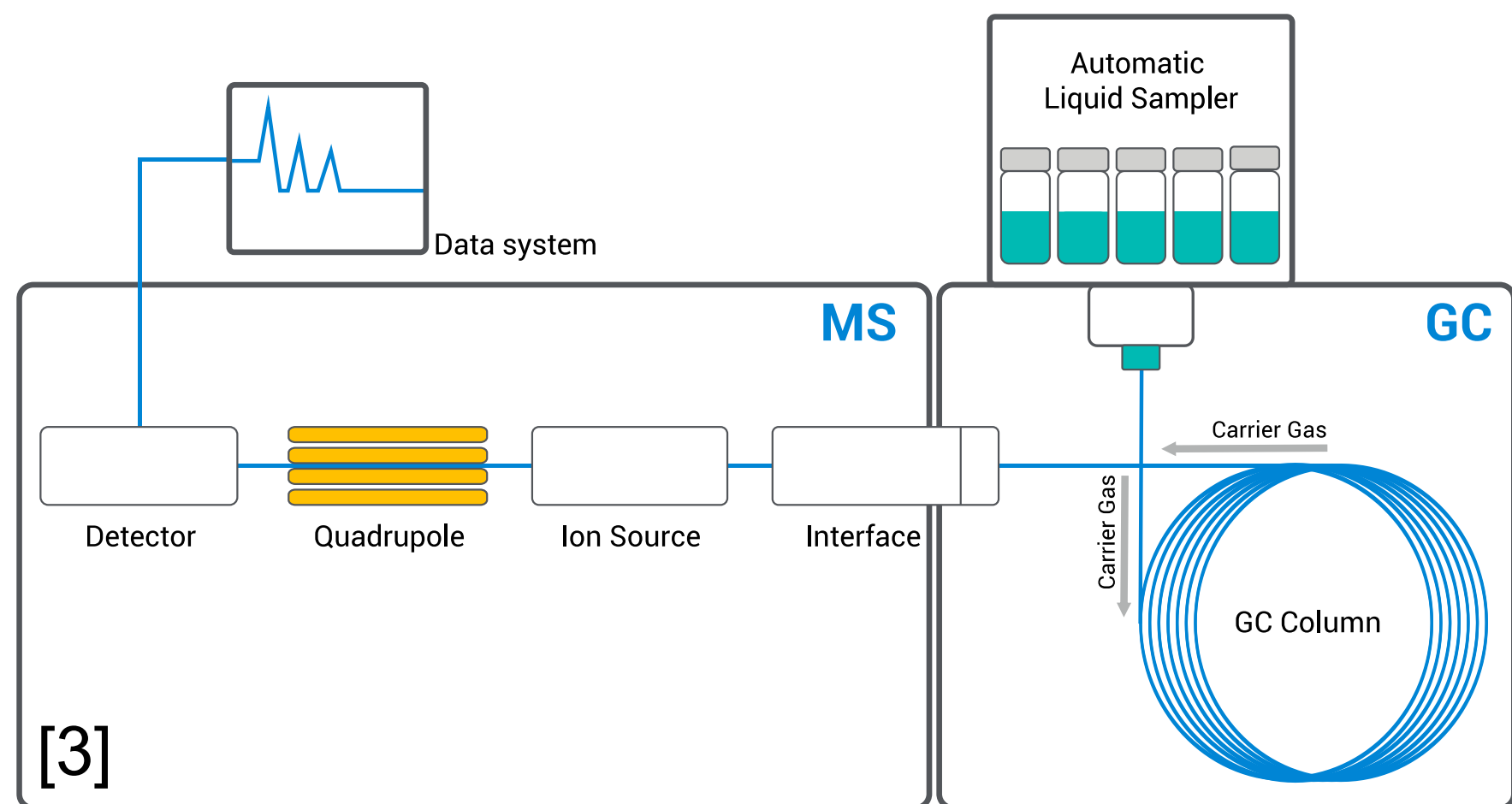
Selected Ion Monitoring (SIM) of 52 m/z:



Ions Measured
(m/z):

79

52



Acknowledgements

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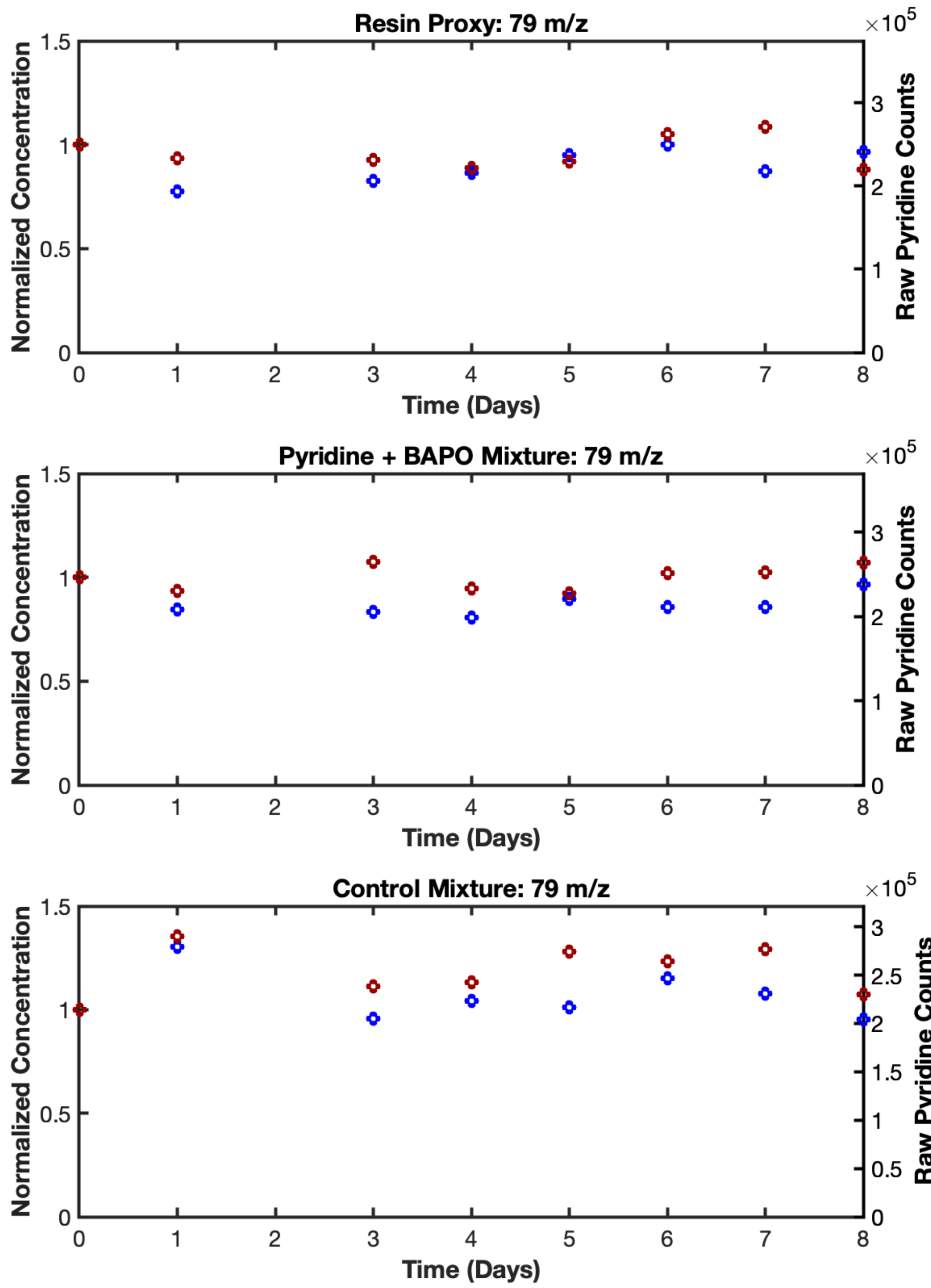
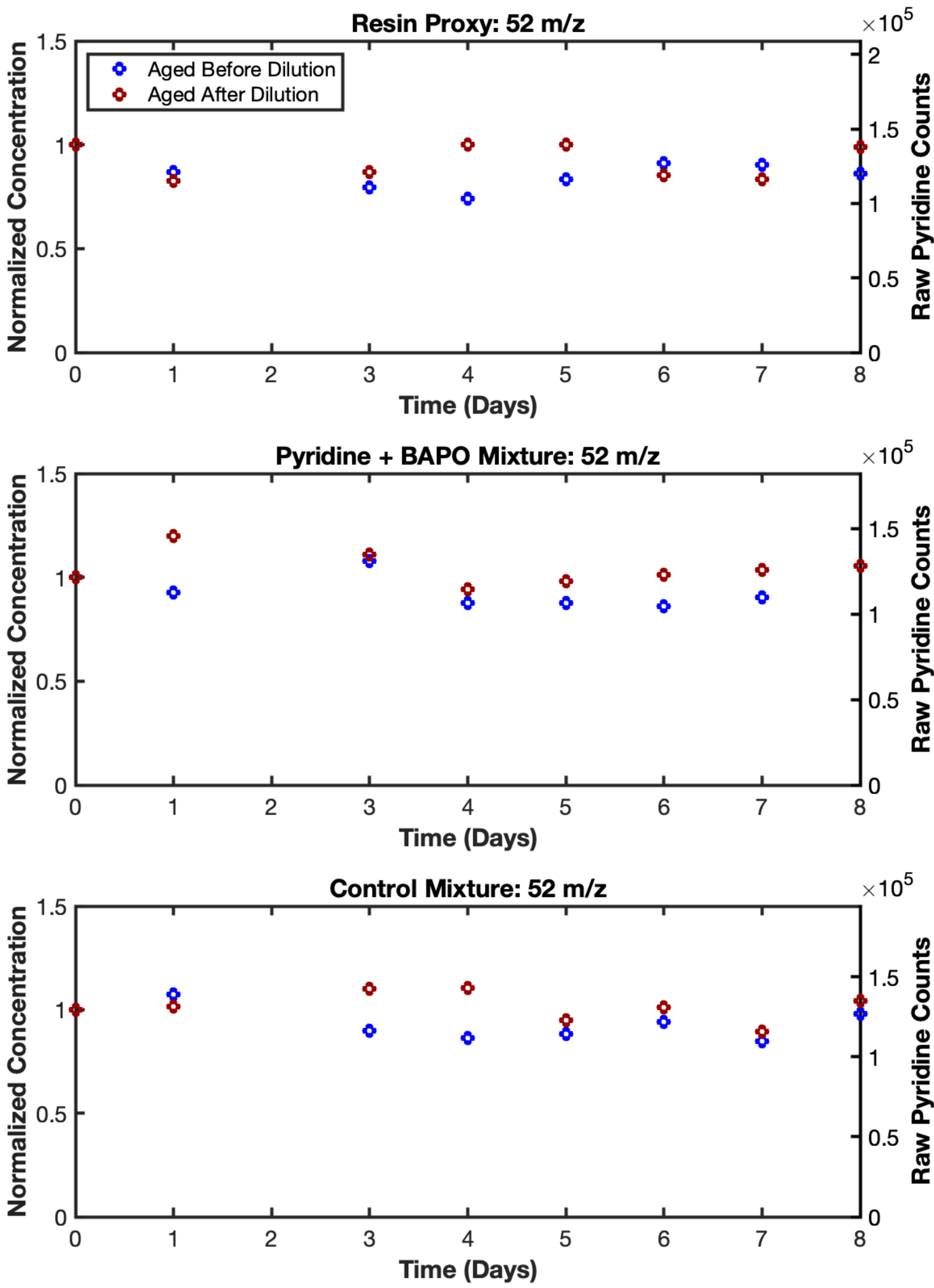
References

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Results



52 m/z (Aged Before Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	-0.0139	-0.0040	-0.0057
95% Confidence	0.0238	0.0303	0.0273

79 m/z (Aged After Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	0.0067	0.0079	0.0028
95% Confidence	0.0431	0.0195	0.0272

52 m/z (Aged After Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	-0.0063	-0.0082	-0.0003
95% Confidence	0.0244	0.0276	0.0292

79 m/z (Aged Before Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	-0.0115	-0.0016	0.0083
95% Confidence	0.0402	0.0238	0.0286

Conclusions

- Pyridine levels in the resin remained stable in all samples for the duration of the testing period
- Pyridine levels in the resin are not significantly different across samples tested
- Resin stability can be attributed to mechanisms other than pyridine degradation
- Measurement error was introduced by instrument saturation and absence of an internal standard and calibration curve. Future work should include these for additional certainty in the results.

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